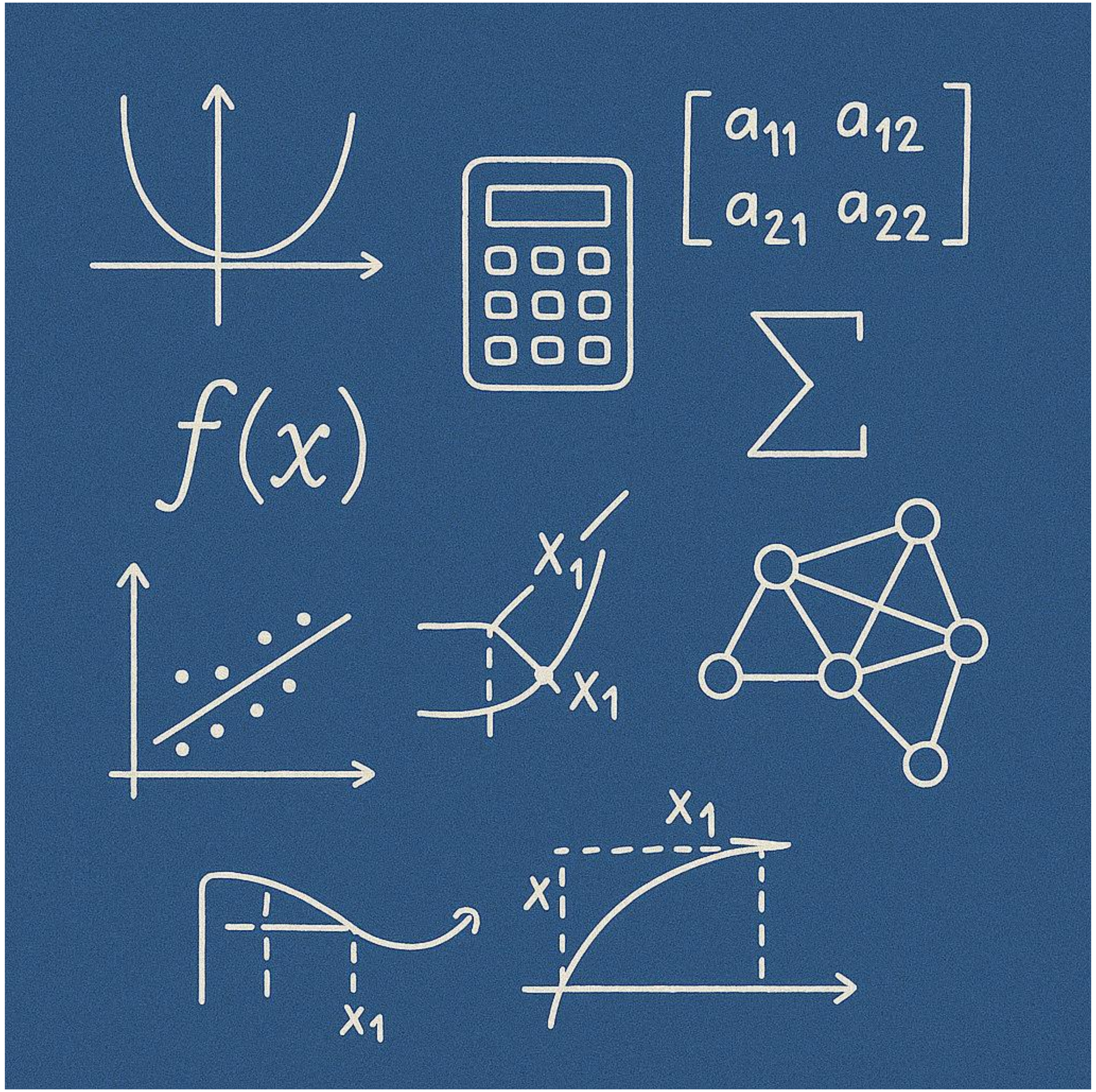
Métodos Numéricos ICCD412

U6 Solución numérica de EDOs

Carlos Ayala T.



Contenido

- Método de Taylor de mayor orden
- Orden
- Runge Kutta

Método de Taylor de mayor orden

Basado en polinomios de Taylor de orden mayor a 1.

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2!} f'(t_i, y(t_i)) + \cdots + \frac{h^n}{n!} f^{(n-1)}(t_i, y(t_i)) + \mathcal{O}(h^{n+1})$$

Calcular derivadas puede ser costoso

$$T^{(n)}(t_i, w_i) = f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i) + \cdots + \frac{h^{n-1}}{n!} f^{(n-1)}(t_i, w_i)$$

$$w_{i+1} = w_i + h \cdot T^{(n)}(t_i, w_i)$$

Método de Taylor de mayor orden

Aproximar, en el intervalo [0,1] con tamaño de paso h = 0.2

$$\frac{dy}{dt} = t + y, \quad y(0) = 1$$

$$t_0 = 0$$

$$\hat{y}_0 = 1$$

$$f(t, y) = t + y$$

$$\frac{d}{dt} f(t, y) = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

$$\frac{\partial f}{\partial t} = 1$$

$$\frac{\partial f}{\partial y} = 1$$

$$\frac{dy}{dt} = f(t, y) = t + y$$

$$f'(t, y) = 1 + 1 \cdot (t + y) = 1 + t + y$$

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2!} f'(t_i, y(t_i)) + \cdots + \frac{h^n}{n!} f^{(n-1)}(t_i, y(t_i)) + \mathcal{O}(h^{n+1})$$

Método de Taylor de mayor orden

Aproximar, en el intervalo [0,1] con tamaño de paso h = 0.2

$$\frac{dy}{dt} = t + y, \quad y(0) = 1$$

$$t_0 = 0$$

$$\hat{y}_0 = 1$$

$$f(t, y) = t + y$$

$$f(t_0, y_0) = 0 + 1 = 1$$

$$f'(t_0, y_0) = 1 + 0 + 1 = 2$$

$$y_1 = 1 + 0.2 \cdot 1 + \frac{0.2^2}{2} \cdot 2 = 1 + 0.2 + 0.04 = 1.24$$

$$f = 0.2 + 1.24 = 1.44, \quad f' = 1 + 0.2 + 1.24 = 2.44$$

$$y_2 = 1.24 + 0.2 \cdot 1.44 + \frac{0.2^2}{2} \cdot 2.44 = 1.24 + 0.288 + 0.0488 = 1.5768$$

Método de Taylor de mayor orden

Aproximar, en el intervalo [0,1] con tamaño de paso h = 0.2

$$\frac{dy}{dt} = t + y, \quad y(0) = 1$$

i	t_i	y_i
0	0.0	1,0000
1	0.2	1,2400
2	0.4	1,5768
3	0.6	2,0317
4	0.8	2,6307
5	1.0	3,4055

Orden

Qué tan rápido disminuye el error al reducir el tamaño del paso h.

Error Local

En un solo paso

$$\mathcal{O}(h^{p+1})$$

Error Global

Acumulado de todos los errores locales

$$\mathcal{O}(h^{p})$$

Mayor orden, más rápido converge la solución numérica a la solución exacta.

Mayor orden implica más cálculos por paso.

Método Runge Kutta

- Resolver EDOs sin necesidad de derivadas de orden superior.
- Combina varias pendientes.
- Alternativa eficiente al método de Taylor de orden superior.

Método Runge Kutta

Cuarto orden

$$
\begin{aligned}
k_1 &= f(t_n, y_n) \\
k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right) \\
k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right) \\
k_4 &= f(t_n + h, y_n + hk_3) \\
y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}
$$

Método Runge Kutta

$$\frac{dy}{dt} = t + y, \quad y(0) = 1, \quad h = 0.2, \quad t \in [0, 1]$$

$$\begin{aligned}
k_1 &= f(t_n, y_n) \\
k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right) \\
k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right) \\
k_4 &= f(t_n + h, y_n + hk_3) \\
y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}$$

$$\begin{aligned}
k_1 &= f(0, 1) = 0 + 1 = 1 \\
k_2 &= f(0.1, 1 + 0.1) = f(0.1, 1.1) = 0.1 + 1.1 = 1.2 \\
k_3 &= f(0.1, 1 + 0.1 \cdot 1.2) = f(0.1, 1.12) = 0.1 + 1.12 = 1.22 \\
k_4 &= f(0.2, 1 + 0.2 \cdot 1.22) = f(0.2, 1.244) = 0.2 + 1.244 = 1.444 \\
y_1 &= 1 + \frac{0.2}{6}(1 + 2(1.2) + 2(1.22) + 1.444) \\
&= 1 + \frac{0.2}{6}(1 + 2.4 + 2.44 + 1.444) \\
&= 1 + \frac{0.2}{6}(7.284) = 1 + 0.243 \\
&\approx 1.2428
\end{aligned}$$

Método Runge Kutta

$$\frac{dy}{dt} = t + y, \quad y(0) = 1, \quad h = 0.2, \quad t \in [0, 1]$$

$$\begin{aligned}
k_1 &= f(0.2, 1.2428) = 0.2 + 1.2428 = 1.4428 \\
k_2 &= f(0.3, 1.2428 + 0.1 \cdot 1.4428) = f(0.3, 1.3871) = 0.3 + 1.3871 = 1.6871 \\
k_3 &= f(0.3, 1.2428 + 0.1 \cdot 1.6871) = f(0.3, 1.4115) = 0.3 + 1.4115 = 1.7115 \\
k_4 &= f(0.4, 1.2428 + 0.2 \cdot 1.7115) = f(0.4, 1.5851) = 0.4 + 1.5851 = 1.9851 \\
y_2 &= 1.2428 + \frac{0.2}{6}(1.4428 + 2(1.6871) + 2(1.7115) + 1.9851) \\
&= 1.2428 + \frac{0.2}{6}(1.4428 + 3.3742 + 3.423 + 1.9851) \\
&= 1.2428 + \frac{0.2}{6}(10.2251) \approx 1.2428 + 0.3408 \\
&\approx 1.5836
\end{aligned}$$

Método Runge Kutta

$$\frac{dy}{dt} = t + y, \quad y(0) = 1, \quad h = 0.2, \quad t \in [0,1]$$

$$\begin{aligned}
k_1 &= f(0.4, 1.5836) = 0.4 + 1.5836 = 1.9836 \\
k_2 &= f(0.5, 1.5836 + 0.1 \cdot 1.9836) = f(0.5, 1.7820) = 0.5 + 1.7820 = 2.2820 \\
k_3 &= f(0.5, 1.5836 + 0.1 \cdot 2.2820) = f(0.5, 1.8118) = 0.5 + 1.8118 = 2.3118 \\
k_4 &= f(0.6, 1.5836 + 0.2 \cdot 2.3118) = f(0.6, 2.0460) = 0.6 + 2.0460 = 2.6460 \\
y_3 &= 1.5836 + \frac{0.2}{6}(1.9836 + 2(2.2820) + 2(2.3118) + 2.6460) \\
&= 1.5836 + \frac{0.2}{6}(1.9836 + 4.564 + 4.6236 + 2.646) \\
&= 1.5836 + \frac{0.2}{6}(13.8172) \approx 1.5836 + 0.4606 \\
&\approx 2.0442
\end{aligned}$$

Método Runge Kutta

$$\frac{dy}{dt} = t + y, \quad y(0) = 1, \quad h = 0.2, \quad t \in [0, 1]$$

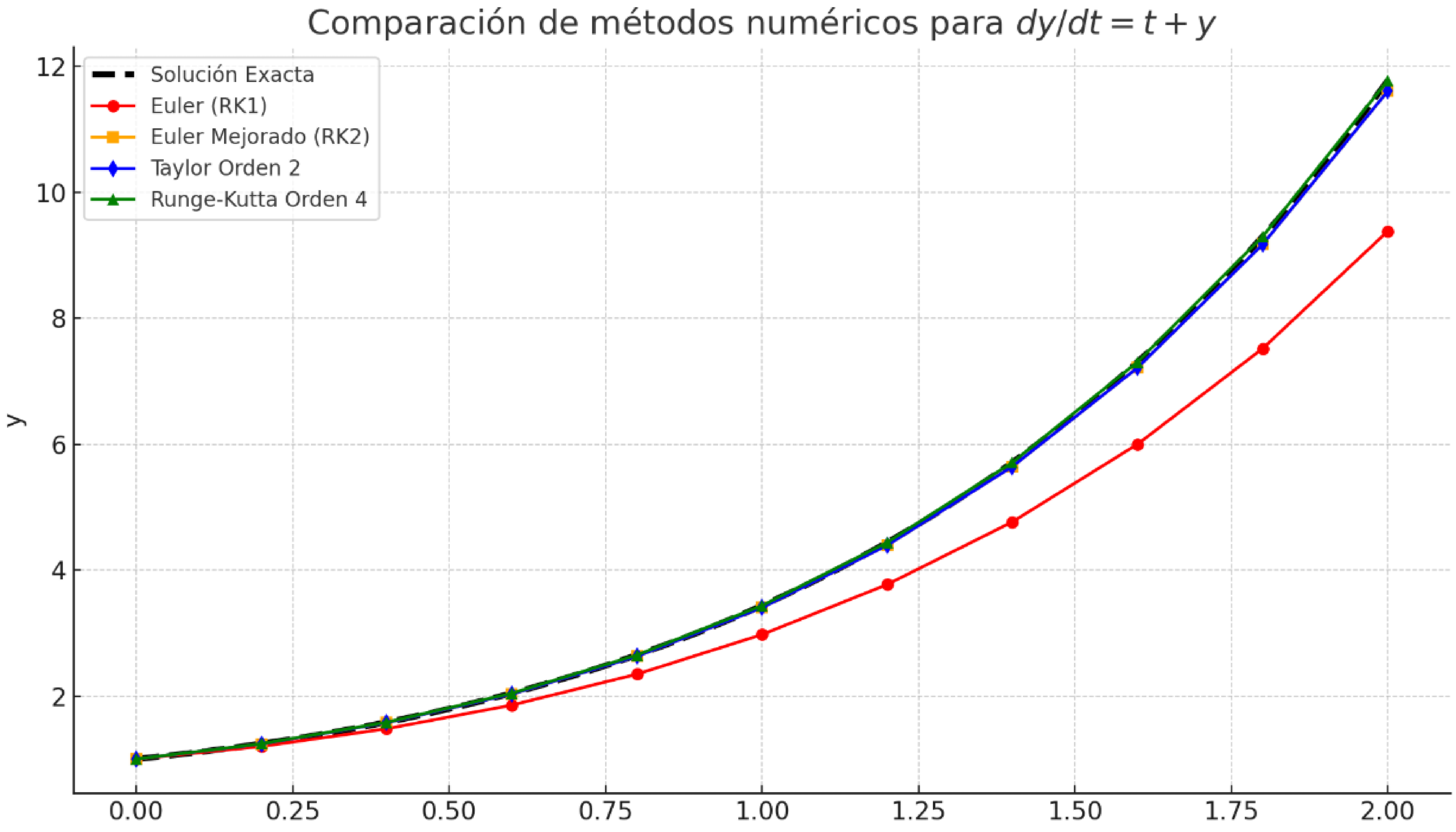
i	t_i	y_i	k_1	k_2	k_3	k_4	y_i+1
0	0.0	1,000	1,000	1,200	1,220	1,444	1,243
1	0.2	1,243	1,443	1,687	1,712	1,985	1,584
2	0.4	1,584	1,984	2,282	2,312	2,646	2,044
3	0.6	2,044	2,644	3,009	3,045	3,453	2,651
4	0.8	2,651	3,451	3,896	3,941	4,439	3,436

Método Runge Kutta

Método	k_1	k_2	k_3	k_4	y_n+1
Euler (RK1)	$f(t_n, y_n)$				$y_{n+1} = y_n + h \cdot k_1$
Euler Mejorado (RK2)	$f(t_n, y_n)$	$f(t_n + h, y_n + h \cdot k_1)$			$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2)$
RK3	$f(t_n, y_n)$	$f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$	$f(t_n + h, y_n - h \cdot k_1 + 2h \cdot k_2)$		$y_{n+1} = y_n + \frac{h}{6}(k_1 + 4k_2 + k_3)$
RK4	$f(t_n, y_n)$	$f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$	$f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$	$f(t_n + h, y_n + h \cdot k_3)$	$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

Runge-Kutta es una generalización del método de Euler.

Método Runge Kutta



Temas a codificar

Método de Taylor de mayor orden

Método de Runge Kutta


THANK YOU
OBRIGADO
gracias
どうも
ARIGATO
hvala
DANK U
takk
MERSI
merci
謝謝
obrigado
danke schön
köszi
PALDIES
muchas gracias
ありがとう
TEŞEKKÜR EDERIM
MOLTE GRAZIE
GO RAIBH MAITH AGAT
danke
grazas
GRAZZI
THANKS
qujan
TAK
благодаря
TAK
どうも
asante
muchas gracias
vielen dank
PALDIES
DANK U
OBRIGADO
mesi
danke
감사합니다
köszi
благодаря
grazie
DZIĘKI
TACK
Gràcies
спасибо
MULTUMESC
多謝
TEŞEKKÜR EDERIM
muchas gracias
obrigado
NA GODE